

$$= \begin{bmatrix} 3 \\ 6 \end{bmatrix} \text{ (Vector addition)}$$

$$b) \begin{bmatrix} 2 & -3 \\ 8 & 0 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 7 \end{bmatrix} = 4 \begin{bmatrix} 2 \\ 8 \\ -5 \end{bmatrix} + 7 \begin{bmatrix} -3 \\ 0 \\ 2 \end{bmatrix} \text{ (By Defn. of } Ax)$$

$$= \begin{bmatrix} 8 \\ 32 \\ -20 \end{bmatrix} + \begin{bmatrix} -21 \\ 0 \\ 14 \end{bmatrix} \text{ (Scalar multiplication)}$$

$$= \begin{bmatrix} -13 \\ 32 \\ -6 \end{bmatrix} \text{ (Vector addition)}$$

Example 2: For v_1, v_2, v_3 in $\mathbb{R}^m$, write the linear combination $3v_1 - 5v_2 + 7v_3$ as matrix times a vector.

Solution: $3v_1 - 5v_2 + 7v_3 = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} \begin{bmatrix} 3 \\ -5 \\ 7 \end{bmatrix}$, where $A = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}$ is a $m \times 3$ matrix

and $x = \begin{bmatrix} 3 \\ -5 \\ 7 \end{bmatrix}$ is a 1×3 column vector. $\square$

Section 1.3 showed how to write a system of linear equations as a vector equation involving a linear combination of vectors. For example, the system

$$\begin{cases} x_1 + 2x_2 - x_3 = 4 \\ -5x_2 + 3x_3 = 1 \end{cases} \quad (1)$$

$$\text{is equivalent to } x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ -5 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix} \quad (2)$$

As in example 2, the linear combination on the left side is a matrix times a vector, so that (2) becomes

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix} \quad (3)$$

Equation (3) has the form $AX=b$. Such an equation is called a **matrix eqn.** to distinguish it from a vector equation such as shown in (2).